Example 7.17

Find the approximate value of $\sqrt[3]{1002}$ (correct to 3 decimal places) using Binomial series.

Solution:

$$\begin{aligned}
\sqrt[3]{1002} &= \sqrt[3]{1000+2} \\
&= \sqrt[3]{1000\left(1+\frac{2}{1000}\right)} \\
&= [(10^3)^{\frac{1}{3}}(1+0.002)^{\frac{1}{3}}] \\
&= 10[1+0.002]^{\frac{1}{3}} \\
&= 10[1+\frac{1}{3}[0.002]+\ldots] \\
&= 10[1+0.00066+\ldots] \\
&= 10[1+0.00066] \\
&= [10+0.0066] \\
&= 10.007
\end{aligned}$$

7.4.2 Exponential series

For all real values of x, $e^x = 1+\frac{x}{1!}+\frac{x^2}{2!}+\frac{x^3}{3!}+\ldots$ is known as exponential series, where e is an irrational number and the value of e to six decimal places is

$e = 2.718282\ldots$ and

$e = 1+\frac{1}{1!}+\frac{1}{2!}+\frac{1}{3!}+\ldots\infty$, $e^{-1}=1-\frac{1}{1!}+\frac{1}{2!}-\frac{1}{3!}+\ldots$

The other exponential expansions are

$e^{-x}= 1-\frac{x}{1!}+\frac{x^2}{2!}-\frac{x^3}{3!}+\ldots$

$\frac{e^x+e^{-x}}{2} = 1+\frac{x^2}{2!}+\frac{x^4}{4!}+\ldots$ and $\frac{e^x-e^{-x}}{2}\ x+\frac{x_3}{3!}+\frac{x^5}{5!}+\ldots$

$\frac{e+e^{-1}}{2} = 1+\frac{1!}{2!}+\frac{1}{4!}+\ldots$ and $\frac{e-e^{-1}}{2}1+\frac{1}{3!}+\frac{1}{5!}+\ldots$

Example 7.18

Show that $\frac{\frac{1}{1!}+\frac{1}{3}+\frac{1}{5!}+\ldots}{\frac{1}{2!}+\frac{1}{4!}+\frac{1}{6!}+\ldots} = \frac{e+1}{e-1}$